

Access to Credit in the United States: Challenges and Solutions for Hispanic and Immigrant Communities

Benedict Guttman-Kenney (Rice University)

Annual Meeting of Financial Education Desks of the Mexican Consulates, 2nd September 2026

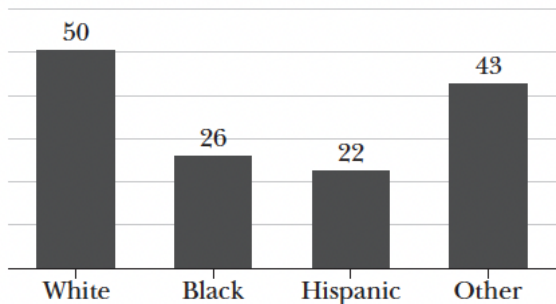
1. Challenges for Minorities in the Financial System
2. Immigration and Credit in the United States
3. Technological Solutions

1. Challenges for Minorities in the Financial System

U.S. Financial Literacy Is Poor, Especially Among Hispanics (Lusardi & Mitchell, 2023)

- Only 43% of U.S. consumers correctly answer all three questions (interest, inflation, risk)

Panel A. Percent of respondents answering all Big Three questions correctly, by race/ethnicity

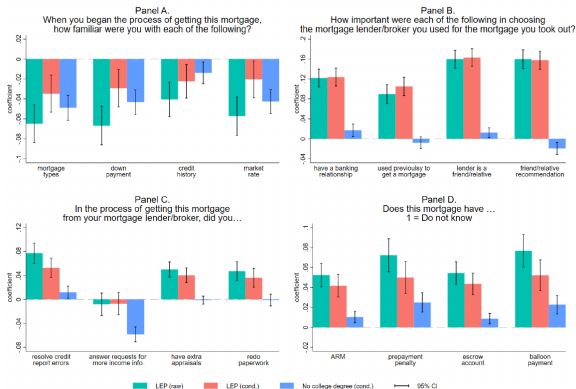


- Low financial literacy correlated with poor financial outcomes

“Language Frictions in Consumer Credit” – Liu (2025)

- $\approx 10\%$ of U.S. mortgage holders have “Limited English Proficiency” (LEP)
- LEP consumers: poorer understanding of mortgages, search less, & pay more

Figure 2. Estimated Differences between LEP and non-LEP Borrowers



“Language Frictions in Consumer Credit” – Liu (2025)

- 2018 regulation provided Spanish-translated mortgage documents to mortgage lenders
 - Later also offered other translations(Federal Housing Finance Agency Language Access Plan: FHFA Mortgage Translations)
- Regulation caused Hispanic LEP to have:
 - Improved financial literacy
 - Increased search for mortgages
 - Lowered mortgage borrowing costs
- Policy increased credit access without increasing lender risk or defaults

Disparities in Credit Outcomes for Minorities

- Minorities pay more for car loans (Butler, Mayer, & Weston, 2023) and often end up with pricier mortgages (Bayer, Ferreira, & Ross, 2018; Gerardi, Willen, & Zhang, 2023)
- Mortgage price gap due to poor lender choice & refinancing less (Bhutta & Hizmo, 2021; Hurtado & Sakong, 2024; Bhutta, Hizmo, & Ringo, 2025)
- Lower minority approval rates for autos loans and mortgages remain (Butler et al., 2023; Bhutta et al., 2025)

Disparities in Credit Outcomes for Minorities

- Minorities pay more for car loans (Butler, Mayer, & Weston, 2023) and often end up with pricier mortgages (Bayer, Ferreira, & Ross, 2018; Gerardi, Willen, & Zhang, 2023)
- Mortgage price gap due to poor lender choice & refinancing less (Bhutta & Hizmo, 2021; Hurtado & Sakong, 2024; Bhutta, Hizmo, & Ringo, 2025)
- Lower minority approval rates for autos loans and mortgages remain (Butler et al., 2023; Bhutta et al., 2025)

What can help minorities to access credit?

- Automation ↓ disparities & ↑ lending (Howell, Kuchler, Snitkof, Stroebe, & Wong, 2024)
- ↓ rates if borrow through credit unions (Shahidinejad, 2026)
- ↑ approvals via minority-owned banks & minority loan officers (Hurtado & Sakong, 2023)

Auto Lenders Impacted By Government Policy Change Towards Immigrants

- Lendbuzz (2025) Pre-IPO Disclosure:

“Changes in immigration patterns, policy or enforcement could affect some of our consumers, including those who may be undocumented immigrants, and consequently impact the performance of our loans, our business”

- Tricolor filed for bankruptcy in 2025 (may be more driven by corporate fraud)

Government Mortgage Policy Towards Minorities & Immigrants Changing

- Regulators revised Equal Credit Opportunity Act (ECOA) and scaled back enforcement.
 - In particular, removed “disparate impact” test
 - Now only evaluate “disparate treatment”
- U.S. Department of Housing and Urban Development removed eligibility for “non-permanent residents” from Federal Housing Administration programs
- Consumer Financial Protection Bureau (CFPB) issued new guidance that lenders consider an applicant’s immigration status in their ability-to-repay requirements

2. Immigration and Credit in the United States

J. Anthony Cookson¹, Benedict Guttman-Kenney², & William Mullins³

¹Penn State University, ²Rice University, ³University of California San Diego

TransUnion (the data provider) has the right to review the research before dissemination to ensure it accurately describes TransUnion data, does not disclose confidential information, and does not contain material it deems to be misleading or false regarding TransUnion, TransUnion's partners, affiliates or customer base, or the consumer lending industry.

- Immigrants to the U.S. bring their labor market experiences (e.g., education records, work history) helping them to navigate U.S. labor markets

- Immigrants to the U.S. bring their labor market experiences (e.g., education records, work history) helping them to navigate U.S. labor markets
- Immigrants start life in the U.S. *without* a U.S. credit report
 - Any foreign credit history generally unobserved to U.S. lenders

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Data

- Representative anonymized sample of U.S. consumer credit reporting data, 2000 to 2024
- Data **matched** with information on whether a consumer is an **immigrant** & **year of immigration**, using sequential Social Security Number (SSNs) assignment

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Data

- Representative anonymized sample of U.S. consumer credit reporting data, 2000 to 2024
- Data **matched** with information on whether a consumer is an **immigrant** & **year of immigration**, using sequential Social Security Number (SSNs) assignment

What We Do

- Describe **selection** of Immigrants compared to Non-Immigrants in their creditworthiness
- Describe **assimilation** in creditworthiness & access to credit over life cycle
- How **dynamics** of life cycle outcomes vary by one year difference in age of immigration (with fixed effects for birth year & ZIP5 geography)

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Data

- U.S. consumer credit reporting data matched with immigration data

What We Do

- Describe **selection** of Immigrants & **assimilation** dynamics over life cycle
 - Immigrants vs. same-age Non-Immigrants & Immigrants arriving at different ages

Immigrants (vs. Non-Immigrants), Immigrants Arriving At Older (vs. Younger) Ages:

- **Positive selection:** More creditworthy (once credit scored)
- Despite better creditworthiness, **lower long-term access** to auto loans & mortgages

Timing of immigration best explains most of our results (“**history dependence**”)

Our Paper Contributes To Labor Economics & Household Finance Literatures

- Prior work shows the large positive contribution of legal immigrants to U.S. economy (e.g., Azoulay et al., 2022; Bernstein et al., 2025)
- We show the **selection** of immigrants based on their credit behaviors, complementing work focusing on labor market outcomes (e.g., Borjas, 1985; Abramitzky & Boustan, 2017)
- We show **assimilation** into the U.S. credit markets over life cycle, complementing work on labor & cultural assimilation (e.g., Borjas, 1985; Abramitzky et al., 2014, 2020, 2021; Lubotsky, 2007; Bleakley & Chin, 2010; Doran, Gelber, & Isen, 2022; Bailey et al., 2022)
- We contribute to the household finance literature that has largely not studied immigrants (e.g., Dobbie et al., 2021; Zillessen, 2022; Bakker et al., 2025; Gorback & Schubert, 2025)

1. Data & Institutional Details
2. Credit Market Entry
3. Creditworthiness
4. Credit Access
5. Mechanisms

Data & Institutional Details

University of Chicago Booth TransUnion Consumer Credit Panel (BTCCP)

- Anonymized data sample of 1 in 10 consumers with U.S. credit reports
- Monthly data from 2000 to 2024
- Individual tradeline data (e.g., details on each account, including opening dates) & consumer-level data (e.g., birth date, credit score, ZIP code)

Gibbs, Guttman-Kenney, Lee, Nelson, van der Klaauw & Wang (2025)

University of Chicago Booth TransUnion Consumer Credit Panel (BTCCP)

- Anonymized data sample of 1 in 10 consumers with U.S. credit reports
- Monthly data from 2000 to 2024
- Individual tradeline data (e.g., details on each account, including opening dates) & consumer-level data (e.g., birth date, credit score, ZIP code)

Gibbs, Guttman-Kenney, Lee, Nelson, van der Klaauw & Wang (2025)

Also Use Public Data To Help Evaluate Mechanisms:

- **American Community Survey (ACS) 5-Year Waves Spanning 2006 to 2023**
- **2022 Survey of Consumer Finances (SCF)**

Classifying Immigrants In The BTCCP

Before 2012, U.S. Social Security Administration assigned first 5-digits of 9-digit Social Security Numbers (SSNs) sequentially within states (Puckett, 2009)

Consumers assigned SSNs as adults are very likely to be (legal) immigrants

(Klopfer & Miller, 2024; Yonker, 2017; Doran, Gelber, & Isen, 2022; Bernstein et al., 2025)

We calculate:

$$SSN\ Age = SSN\ Year - Birth\ Year$$

“Immigrants”: assigned SSN at age 21+. **“Non-Immigrants”**: assigned SSN at age < 21

Classifying Immigrants In The BTCCP

Before 2012, U.S. Social Security Administration assigned first 5-digits of 9-digit Social Security Numbers (SSNs) sequentially within states (Puckett, 2009)

Consumers assigned SSNs as adults are very likely to be (legal) immigrants

(Klopfer & Miller, 2024; Yonker, 2017; Doran, Gelber, & Isen, 2022; Bernstein et al., 2025)

We calculate:

$$SSN\ Age = SSN\ Year - Birth\ Year$$

“Immigrants”: assigned SSN at age 21+. **“Non-Immigrants”**: assigned SSN at age < 21

Restrict To Birth Years 1975 to 1987 With SSN Ages 0 to 29:

6,299,287 consumers – 384,396 (6.1%) “Immigrants” (SSN Age 21 to 29)

Follow **credit outcomes** in BTCCP for **ages 18 to 40**

- **Information frictions** may reduce immigrant credit access:
 1. Immigrants have **no credit report on arrival**
 - **Lender's prior:** Consumer with no U.S. credit report is **high credit risk**
 2. After entering the credit system, immigrants have a **shorter credit history**
 - Length of credit report has positive impact on credit scores ($\approx 15\%$ FICO)

- **Information frictions** may reduce immigrant credit access:
 1. Immigrants have **no credit report on arrival**
 - **Lender's prior:** Consumer with no U.S. credit report is **high credit risk**
 2. After entering the credit system, immigrants have a **shorter credit history**
 - Length of credit report has positive impact on credit scores ($\approx 15\%$ FICO)
- Different cultural **tastes** & informational barriers for consumers
 - Differences in familiarity & willingness to use debt, understanding of how to build credit score
- Lenders use credit scores (+ other information) in decisions
 - Can use immigration status, can't use protected characteristics (e.g., national origin, race) under Equal Credit Opportunity Act (ECOA)

Credit Market Entry

Main Cross-Sectional Regression Specifications

β_1 : average difference for Immigrants vs. Non-Immigrants from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \quad (1)$$

– Includes fixed effects: birth year ($\mu_{b(i)}$) and geography based on first ZIP5 ($\eta_{g(i)}$)

Main Cross-Sectional Regression Specifications

β_1 : average difference for Immigrants vs. Non-Immigrants from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \quad (1)$$

– Includes fixed effects: birth year ($\mu_{b(i)}$) and geography based on first ZIP5 ($\eta_{g(i)}$)

β_2 : marginal difference for SSN Age one year later, from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \quad (2)$$

– Cluster standard errors by birth year $\times$ ZIP2

Main Cross-Sectional Regression Specifications

β_1 : average difference for Immigrants vs. Non-Immigrants from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \quad (1)$$

– Includes fixed effects: birth year ($\mu_{b(i)}$) and geography based on first ZIP5 ($\eta_{g(i)}$)

β_2 : marginal difference for SSN Age one year later, from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \quad (2)$$

– Cluster standard errors by birth year $\times$ ZIP2

Results robust to including additional F.E.:

– Last ZIP5, Longest-held ZIP5, Number of ZIP5 (geographic mobility proxy), & credit score.

Also robust to restricting to consumers who have not emigrated by 2024.

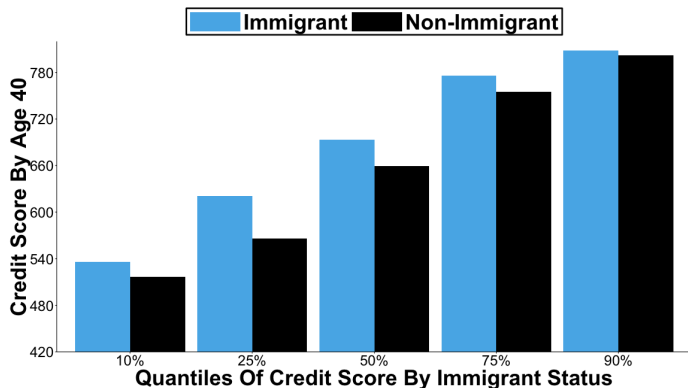
Immigrants, & Older SSN Age Immigrants, Enter Credit Market Later Than Non-Immigrants & Younger SSN Age Immigrants

	Age at First Credit Report	
	(1)	(2)
Immigrant	5.53*** (0.02)	2.29*** (0.02)
SSN Age		0.75*** (0.01)
F.E. Birth Year	X	X
F.E. First Zip5	X	X
R^2	0.193	0.209
Mean, SSN Age <21	20.19	20.19

$N = 6,297,340$, *** 0.5%, ** 1%, * 5% statistical significance

Creditworthiness

Positive Selection of Immigrants (vs. Non-Immigrants) In Credit Scores

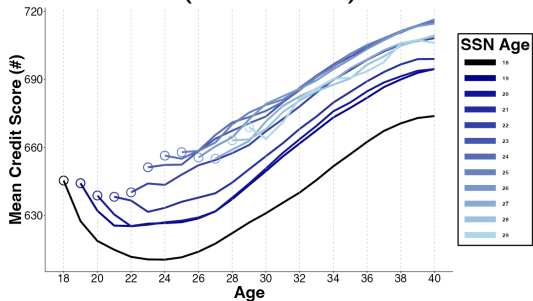


- Credit scores measure financial behavior, distinct from labor market behaviors
- Income & labor market behaviors are not credit score inputs (Gibbs et al., 2025)
- Credit scores are weakly correlated with income (Berman, Ionescu, & Li, 2026)

Immigrants (vs. Non-Immigrants) +26.7 Credit Score By Age 30

Credit Score: +17.2 points immigrant (vs. non-immigrant) + 2.3 for each SSN Age

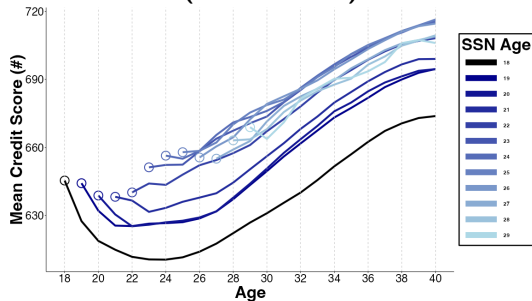
A. Mean Credit Score (Once Scored)



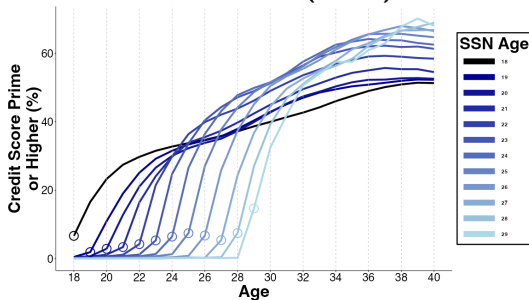
Immigrants (vs. Non-Immigrants) +26.7 Credit Score By Age 30

Credit Score: +17.2 points immigrant (vs. non-immigrant) + 2.3 for each SSN Age

A. Mean Credit Score
(Once Scored)



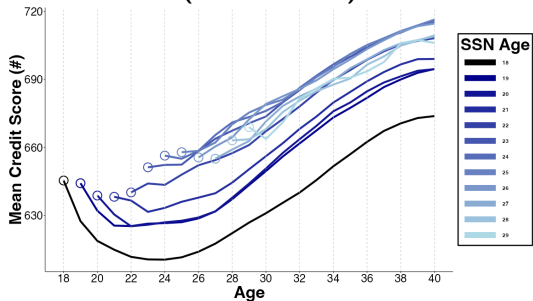
B. Prime or Higher
Credit Score (660+)



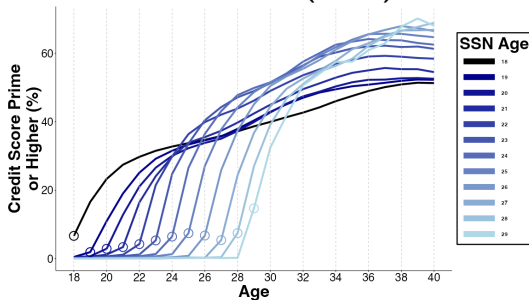
Immigrants (vs. Non-Immigrants) +26.7 Credit Score By Age 30

Credit Score: +17.2 points immigrant (vs. non-immigrant) + 2.3 for each SSN Age

A. Mean Credit Score
(Once Scored)



B. Prime or Higher
Credit Score (660+)

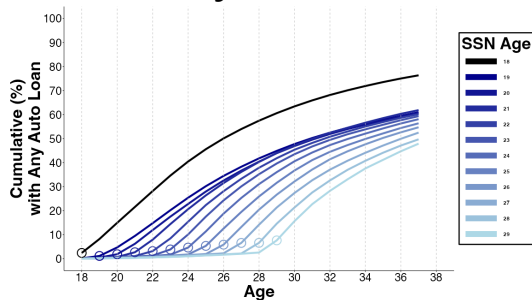


- Immigrants more creditworthy with ↓ delinquency rates & ↓ credit card utilization
- ↓ 5.5% delinquency rates conditional on credit score

Credit Access

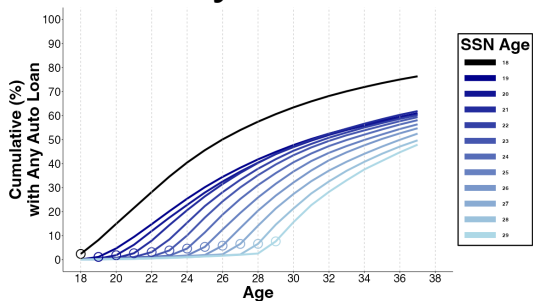
Immigrants Persistently Less Likely To Hold Auto Loans & Mortgages

A. Any Auto Loan

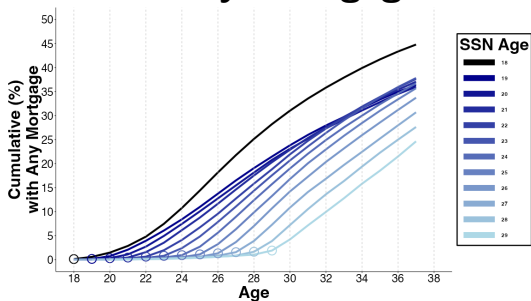


Immigrants Persistently Less Likely To Hold Auto Loans & Mortgages

A. Any Auto Loan



B. Any Mortgage



By Age 37, Immigrants 17% (16%) Less Likely To Have Auto Loan (Mortgage)

↓ Auto Loan & Mortgage Use For Later In Life Immigrants

Dep. Var: By Age 37, has ...	<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-15.66*** (0.18)	-8.69*** (0.23)	-8.91*** (0.17)	-1.05*** (0.23)	-1.35*** (0.08)	-0.37*** (0.11)
SSN Age		-1.62*** (0.04)		-1.83*** (0.04)		-0.23 (0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R ²	0.041	0.042	0.063	0.063	0.024	0.024
Mean, SSN Age <21	76.06	77.06	44.63	44.63	94.12	94.12

$N = 6,297,340$, *** 0.5%, ** 1%, * 5% statistical significance

Results robust to controlling for credit score, additional F.E., & restricting to non-emigrants

“Paired Cohorts”: Compare Pairs of Immigrants With Same Birth Year Where Focal Cohort Arrives in America One Year Later in Life

$$Y_{k(p),t} = \sum_{\tau=-1}^{+13} \delta_{\tau} (\mathbb{I}\{\text{focal cohort}\}_{k(p)} \times \mathbb{I}\{t = \tau\}) + \pi_{k(p)} + \nu_{p,t} + \epsilon_{k(p),t} \quad (3)$$

- 68 pairs $p \in \{k', k''\}$ with same birth year
Focal Cohort (k'): $SSN\ Age_{k'}$
Control Cohort (k''): $SSN\ Age_{k'} - 1$
- 2,176 cohort-by-year observations $(k(p), t)$

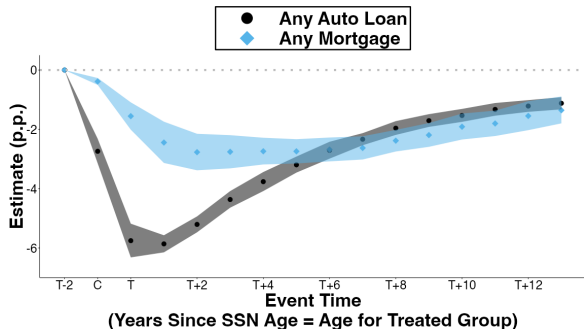
“Paired Cohorts”: Compare Pairs of Immigrants With Same Birth Year Where Focal Cohort Arrives in America One Year Later in Life

$$Y_{k(p),t} = \sum_{\tau=-1}^{+13} \delta_{\tau} (\mathbb{I}\{\text{focal cohort}\}_{k(p)} \times \mathbb{I}\{t = \tau\}) + \pi_{k(p)} + \nu_{p,t} + \epsilon_{k(p),t} \quad (3)$$

- 68 pairs $p \in \{k', k''\}$ with same birth year
Focal Cohort (k'): $SSN\ Age_{k'}$
Control Cohort (k''): $SSN\ Age_{k'} - 1$
- 2,176 cohort-by-year observations ($k(p), t$)

Results

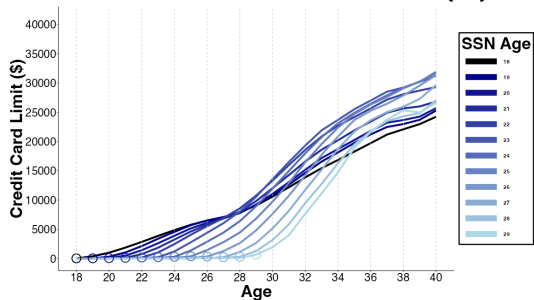
- 13 years post-immigration, one year later-in-life immigrants have:
 - 1.12 p.p. Pr(auto loan)**
(vs. 58.1% control mean)
 - 1.36 p.p. Pr(mortgage)**
(vs. 35.2% control mean)



95% C.I.: wild cluster bootstrapping by birth year

Immigrants' Credit Card Limits Are Lower Than Non-Immigrants

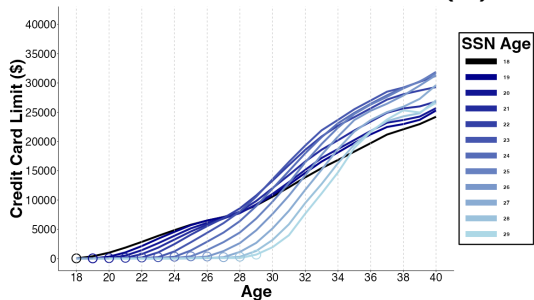
A. Credit Card Limits (\$)



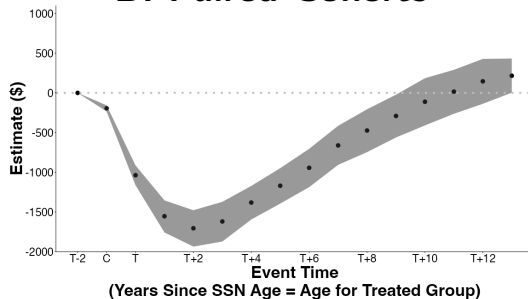
Consistent patterns for:
number of credit cards
limit per credit card

Immigrants' Credit Card Limits Are Lower Than Non-Immigrants

A. Credit Card Limits (\$)



B. Paired Cohorts



Mechanisms

What Mechanisms Explain?

- Less Credit For More Creditworthy Immigrants
- Less Credit For Later In Life Immigrants

~~Creditworthiness (BTCCP)~~

~~Emigration / Return Migration (BTCCP)~~

Credit Demand & Tastes (BTCCP, ACS, SCF)

History Dependence (BTCCP, ACS)

- Credit History Dependence: Financial friction...
 - no U.S. credit report
 - shorter credit history
- Non-Credit History Dependence: less time in U.S. means...
 - fewer years of U.S. earnings
 - less far along American life cycle

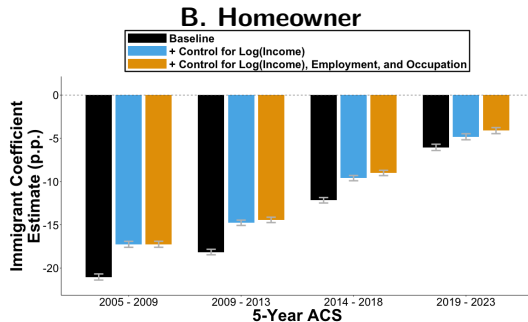
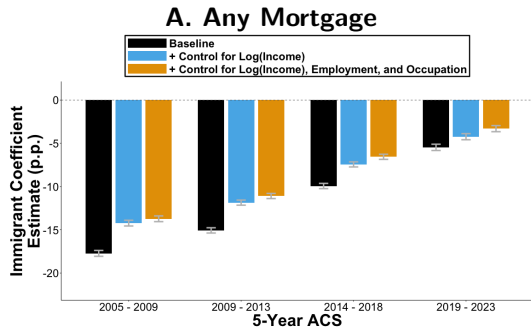
Immigrants Are 19% More Averse To Purchasing Cars With Debt (SCF, 2022)

Dep Var:	<u>Auto Debt Attitude</u>		<u>Any Auto Debt</u>		<u>Any Auto Debt</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-15.41*** (4.58)	-23.02*** (7.25)	-4.27 (5.08)	-0.21 (5.01)	-11.14 (8.27)	-5.10 (7.87)
Immigration Age		0.98 (0.62)			0.88 (0.76)	0.63 (0.69)
Auto Debt Attitude				26.34*** (3.23)		26.24*** (3.24)
Demographic Controls	X	X	X	X	X	X
R^2	0.049	0.050	0.063	0.106	0.063	0.106
Mean, Immigration Age <21	79.58	79.58	47.14	47.14	47.14	47.14

$N = 1,813$, *** 0.5%, ** 1%, * 5%, + 10% statistical significance

No clear evidence of more general negative immigrant attitude to debt

Differences in Mortgage & Homeownership Remain After Controlling for Income & Demographics (5-year ACS waves)



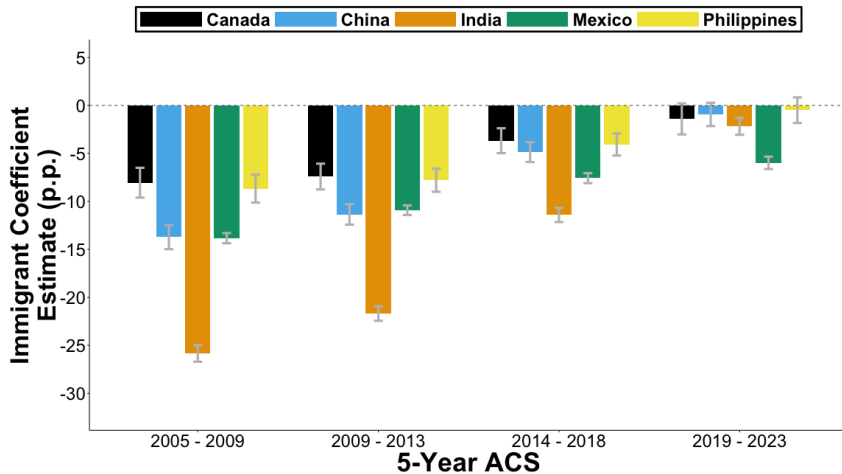
Immigrating one year later in life leads to persistently lower mortgage use & homeownership

Average immigrant vs. non-immigrant difference in car ownership disappears over time

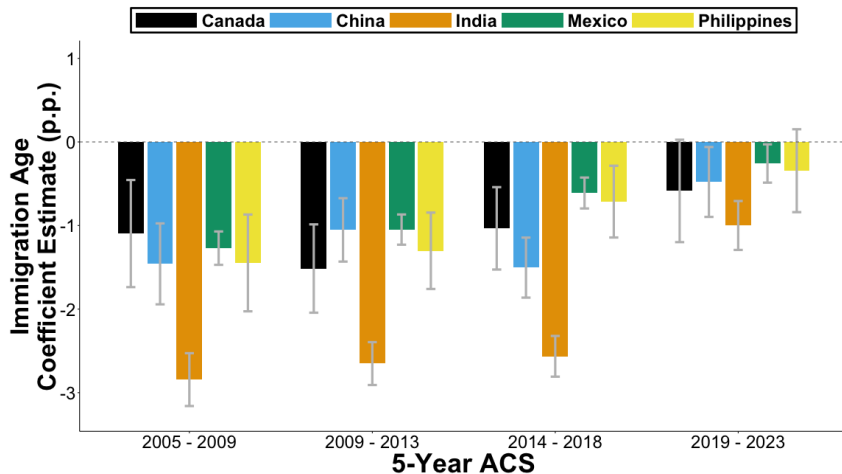
All regressions include fixed effects for...

birth year, geographic area, sex, education levels, race, Hispanic, & survey year

Mexican Immigrants Less Likely To Hold Mortgages



Lower Mortgage Use For Mexican Immigrants Who Immigrate Later In Life



How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Study **selection & assimilation dynamics** over life cycle of Immigrants vs. Non-Immigrants
- exploit variation in age of immigration in credit data matched with immigration information

Immigrants vs. same age Non-Immigrants & Immigrants arriving at older ages:

- **Positively selected:** more creditworthy (once scored)
- Despite better creditworthiness, **lower long-term access** to auto loans & mortgages
 - Lower mortgage use is largely explained by immigration timing
 - Lower auto loan use also due to auto loan-specific debt aversion
 - Not explained by emigration, income, or more general debt aversion
- *Timing* of immigration best explains most of our results (“**history dependence**”)

3. Technological Solutions

Non-Portability Of Credit History Across National Boundaries May Prevent Profitable Lending. Recent Market Solutions?

American Express Global Card Relationship

Global backing for Card Members,
wherever you move

Global Card Relationship, our Card Member-only service, makes it seamless for Card Members to apply for a new American Express® Card in select countries.

Nova Credit

International credit data

Credit Passport® gives you access to foreign consumer-permissioned credit data that's translated to a local equivalent score to serve the 2 million immigrants that arrive in the United States each year and 280 million immigrants worldwide.

Equifax Canada Global Consumer Credit File

Global Consumer Credit File

Helping you grow and retain your clients who are newcomers to Canada

Global Consumer Credit File provides lenders with a seamless and secure platform that connects with global credit bureaus and enables accurate and informed decisioning through robust data.

FinTech lenders (e.g., Lendbuzz, America Mortgages, MPower Financing)

“Alternative” Data Increasingly Used By (FinTech) Lenders To Evaluate Risk Especially For Consumers With Little/No Credit History

- Grocery data (Lee, Yang, & Anderson, 2025; 2026)
- Digital footprints & transactions (Berg, Burg, Gombovic, & Puri, 2020)
e.g., RappiCard in Mexico (Chioda, Gertler, Higgins, & Medina, 2026)
- Education, employment, and device information
e.g., Upstart loans in USA (Di Maggio & Ratnadiwakara, 2026)
- Cashflow data from checking accounts (Plaid, Prism, NovaCredit, Yodlee)
e.g., Arro credit card in USA (Guttman-Kenney & Lee, work-in-progress)

“Alternative” Data Increasingly Used By (FinTech) Lenders To Evaluate Risk Especially For Consumers With Little/No Credit History

- Grocery data (Lee, Yang, & Anderson, 2025; 2026)
- Digital footprints & transactions (Berg, Burg, Gombovic, & Puri, 2020)
e.g., RappiCard in Mexico (Chioda, Gertler, Higgins, & Medina, 2026)
- Education, employment, and device information
e.g., Upstart loans in USA (Di Maggio & Ratnadiwakara, 2026)
- Cashflow data from checking accounts (Plaid, Prism, NovaCredit, Yodlee)
e.g., Arro credit card in USA (Guttman-Kenney & Lee, work-in-progress)
- Machine learning & LLMs may expand lending but adversely affect minorities
(Bartlett, Morse, Stanton, & Wallace, 2022; Fuster, Goldsmith-Pinkham, Ramadorai, & Walther, 2022; Blattner & Nelson, 2026; Bowen, Price, Stein, & Yang, 2025)

- Minorities & immigrants face many challenges accessing finance
 - Lower credit access for immigrants despite higher credit scores
- Groups face more challenges given recent policy changes
- Technological solutions may partially mitigate challenges by
 - Providing data to demonstrate creditworthiness
 - Removing human biases from lending decisions

Thank You. Questions?



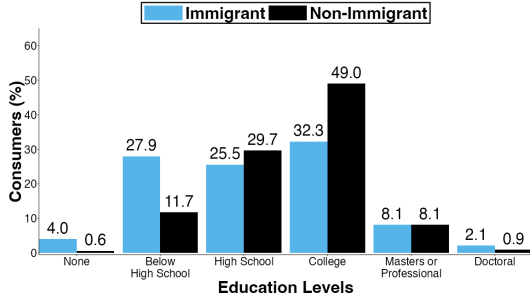
www.benedictgk.com



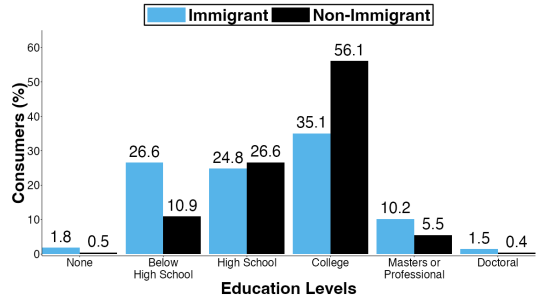
benedictgk@rice.edu

Distribution of Education in 5-year ACS (2010) 🇺🇸

A. All

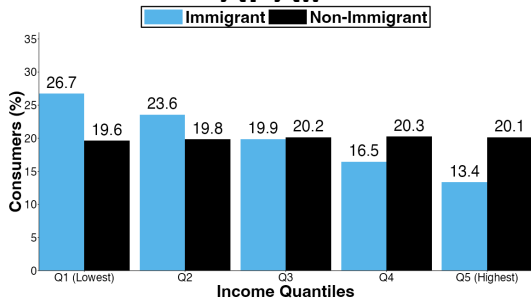


B. Restricted Sample

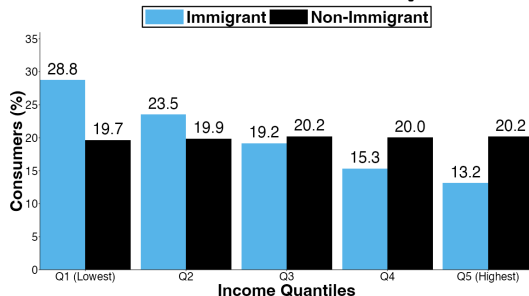


Distribution of Income in 5-year ACS (2010) 🚀

A. All



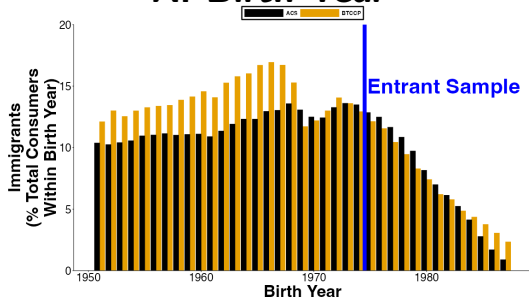
B. Restricted Sample



Immigrant Classification Aligns With American Community Survey (ACS) 🚀

Distribution of Immigrants by...

A. Birth Year



B. Immigration Age



Distribution of immigrants aligns by geography.

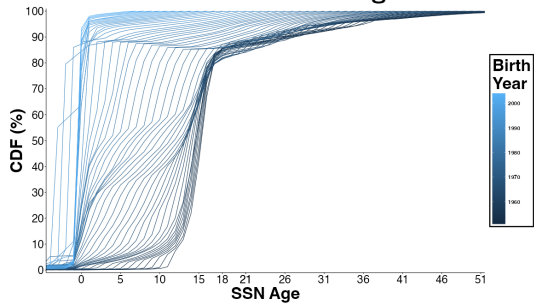
Immigrants also more geographically mobile than non-immigrants.

“Entrant Sample”: Restrict To Birth Years 1975 - 1987 To Observe U.S. Entry At Ages 21 To 29 & Credit Outcomes For Ages 18 To 40 🍷

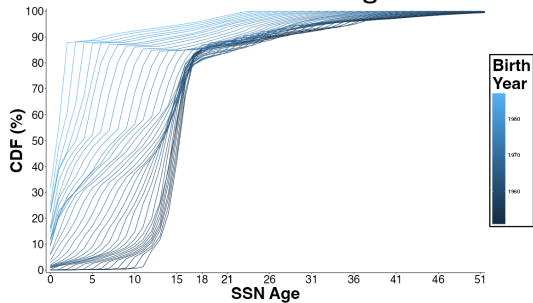
TransUnion Consumers	
... born 1951 to 2004	31,869,445
... with a SSN in TransUnion	25,237,917
 Clean Sample	 19,019,191
... SSN Age <21	16,793,939
... SSN Age 21+	2,225,252 (11.70%)
... Immigrant Cohorts (Birth Year x SSN Year)	775
 Entrant Sample (SSN Age < 30)	 6,299,287
... SSN Age < 21	5,914,891
... SSN Age 21-29	384,396 (6.10%)
... Immigrant Cohorts (Birth Year x SSN Year)	102

SSN Ages By Birth Year

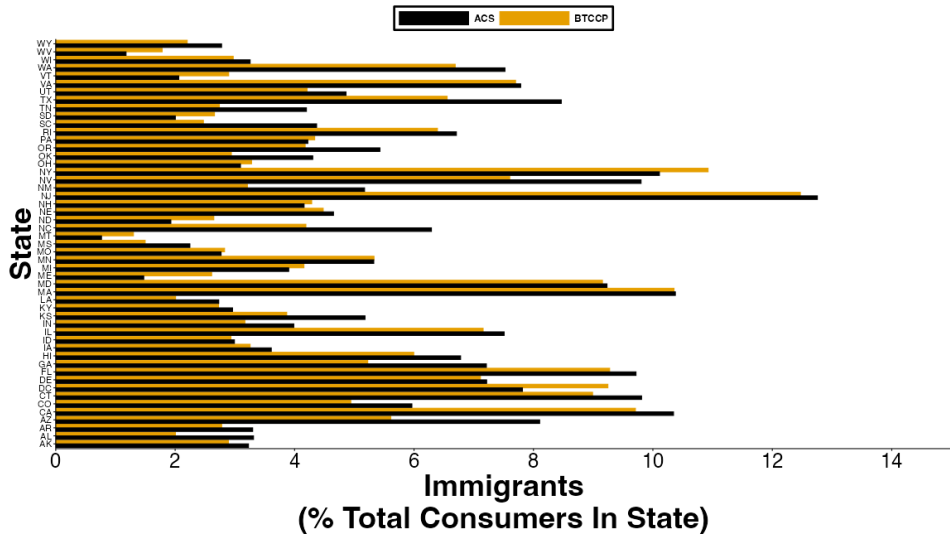
A. Before Cleaning



B. After Cleaning

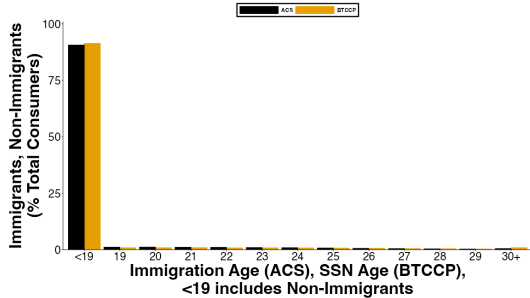


Geographic Comparison of Immigrants: ACS vs. BTCCP 🚀

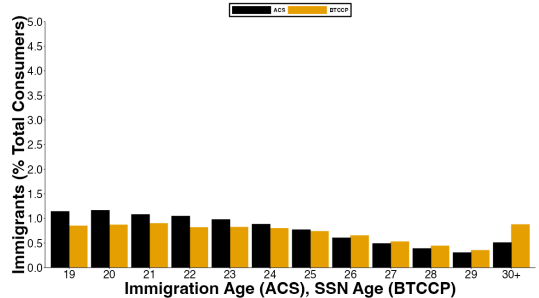


Immigrant Classification in Entrants Sample: ACS vs. BTCCP 🗺️

A. Non-Immigrants and Immigrants by Immigration Age

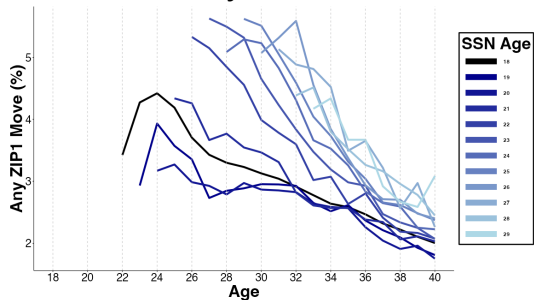


B. Immigrants by Immigration Age

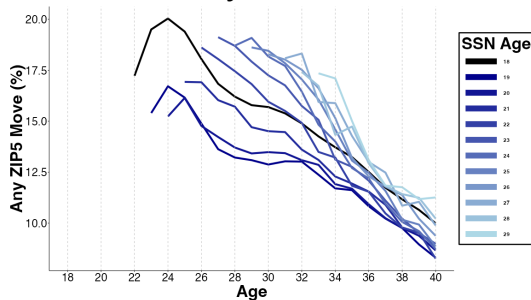


Geographic Mobility 🚀

A. Any ZIP1 Move



B. Any ZIP5 Move



Consumers By SSN Age

SSN Age	Clean Sample	Entrant Sample
<19	16,455,525	5,805,684
19	182,481	53,950
20	155,933	55,257
21	155,743	57,114
22	149,210	51,894
23	153,427	52,314
24	157,685	50,705
25	156,152	46,924
26	150,045	41,481
27	140,610	33,525
28	130,243	28,040
29	119,788	22,399
30+	912,349	55,704

Sample Frame and Timing of SSN Assignment

Calendar year	Birth year												
	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987
1993	≤18												
1994	19	≤18											
1995	20	19											
1996	21	20	≤18										
1997	22	21	20	≤18									
1998	23	22	21	20	19	≤18							
1999	24	23	22	21	20	19	≤18						
2000	25	24	23	22	21	20	19	≤18					
2001	26	25	24	23	22	21	20	19	≤18				
2002	27	26	25	24	23	22	21	20	19	≤18			
2003	28	27	26	25	24	23	22	21	20	19	≤18		
2004	29	28	27	26	25	24	23	22	21	20	19	≤18	
2005	30	29	28	27	26	25	24	23	22	21	20	19	≤18
2006	31	30	29	28	27	26	25	24	23	22	21	20	19
2007	32	31	30	29	28	27	26	25	24	23	22	21	20
2008	33	32	31	30	29	28	27	26	25	24	23	22	21
2009	34	33	32	31	30	29	28	27	26	25	24	23	22
2010	35	34	33	32	31	30	29	28	27	26	25	24	23
2011	36	35	34	33	32	31	30	29	28	27	26	25	24
2012	37	36	35	34	33	32	31	30	29	28	27	26	25
2013	38	37	36	35	34	33	32	31	30	29	28	27	26
2014	39	38	37	36	35	34	33	32	31	30	29	28	27
2015	40	39	38	37	36	35	34	33	32	31	30	29	28
2016	41	40	39	38	37	36	35	34	33	32	31	30	29
2017	42	41	40	39	38	37	36	35	34	33	32	31	30
2018	43	42	41	40	39	38	37	36	35	34	33	32	31
2019	44	43	42	41	40	39	38	37	36	35	34	33	32
2020	45	44	43	42	41	40	39	38	37	36	35	34	33
2021	46	45	44	43	42	41	40	39	38	37	36	35	34
2022	47	46	45	44	43	42	41	40	39	38	37	36	35
2023	48	47	46	45	44	43	42	41	40	39	38	37	36
2024	49	48	47	46	45	44	43	42	41	40	39	38	37

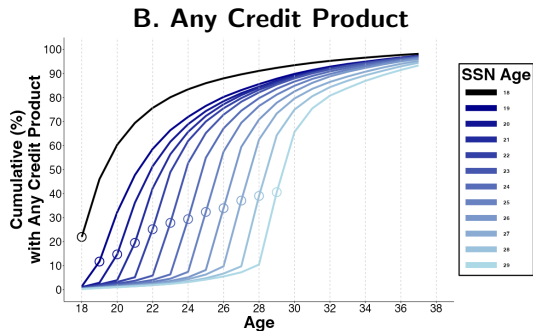
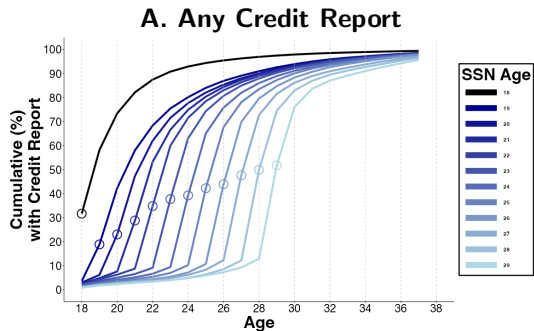
Age at SSN
assignment

Years of
data in our
sample

Summary Statistics

	<u>Non-Immigrants</u> (SSN Age < 21)		<u>Immigrants</u> (SSN Age 21 to 29)	
	Mean	N	Mean	N
Credit Report				
Age at First Credit Report	20.19	5,914,891	26.09	384,396
Credit Score				
Any Score by Age 30 (%)	96.26	5,914,891	81.63	384,396
Credit Score by Age 30	624.2	5,693,584	658.3	313,799
Any Score by Age 40 (%)	99.35	5,914,891	97.94	384,396
Credit Score by Age 40	658.1	5,876,224	687.5	376,461
Auto Loan				
Any Auto Loan (%)	81.14	5,914,891	65.62	384,396
Age at First Auto Loan	26.13	4,799,057	30.59	252,243
Mortgage				
Any Mortgage (%)	50.28	5,914,891	44.42	384,396
Age at First Mortgage	29.53	2,973,846	32.88	170,754
Credit Card				
Any Credit Card (%)	96.87	5,914,891	97.39	384,396
Age at First Credit Card	22.33	5,730,022	27.20	374,365
Credit Card Limits at Age 30	\$11,492	5,914,891	\$10,211	384,396
Credit Card Limits at Age 40	\$22,112	4,579,356	\$26,608	342,258

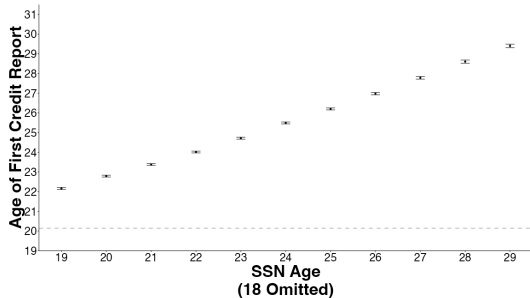
Age At First Credit Report and First Credit Product By SSN Age Cohort



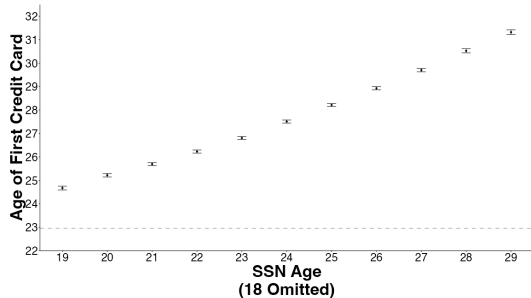
Dep Var: Age at First...	Credit Report		Credit Product	
	(1)	(2)	(3)	(4)
Immigrant	5.53*** (0.02)	2.29*** (0.02)	5.20*** (0.02)	2.07*** (0.03)
SSN Age		0.75*** (0.01)		0.73*** (0.01)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.193	0.209	0.118	0.126
N	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	20.19	20.19	21.56	21.56

Age At First Credit Report & Credit Card 🚀

A. Age at First Credit Report



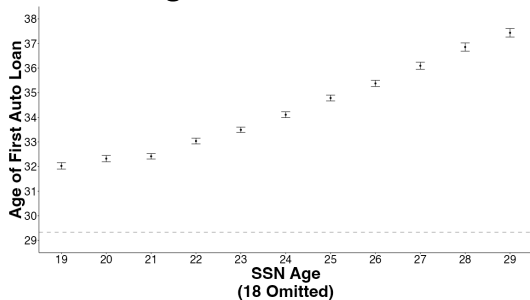
B. Age at First Credit Card



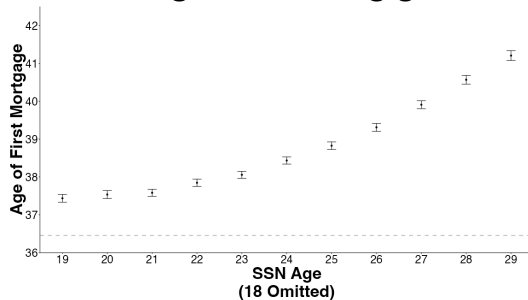
$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i$$

Age At First Auto Loan and Mortgage

A. Age at First Auto Loan



B. Age at First Mortgage

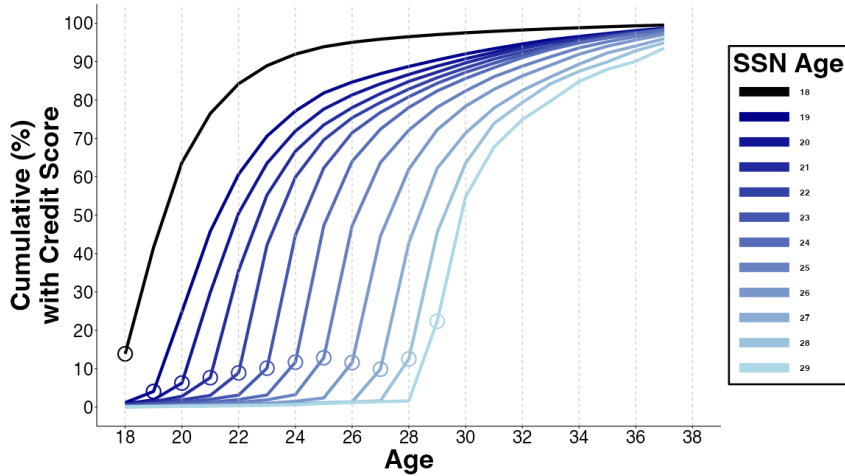


$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i$$

Credit Entry With Additional F.E.

Dep Var: Age at First...	Credit Report		Credit Product		Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	5.15*** (0.02)	2.07*** (0.02)	4.73*** (0.02)	1.81*** (0.03)	3.88*** (0.04)	1.60*** (0.04)	1.72*** (0.03)	0.03 (0.04)	4.21*** (0.02)	1.49*** (0.03)
SSN Age		0.72*** (0.01)		0.68*** (0.01)		0.53*** (0.01)		0.39*** (0.01)		0.63*** (0.01)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X	X	X
R^2	0.260	0.275	0.200	0.207	0.180	0.181	0.158	0.159	0.173	0.176
N	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	20.19	20.19	21.56	21.56	29.40	29.40	36.49	36.49	22.99	22.99

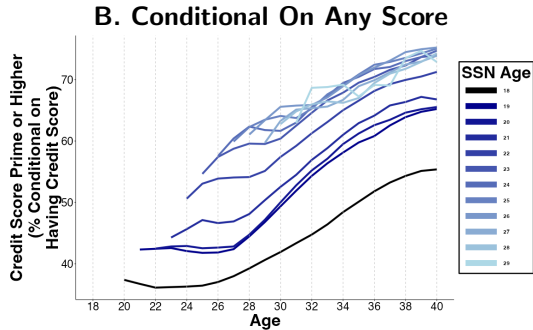
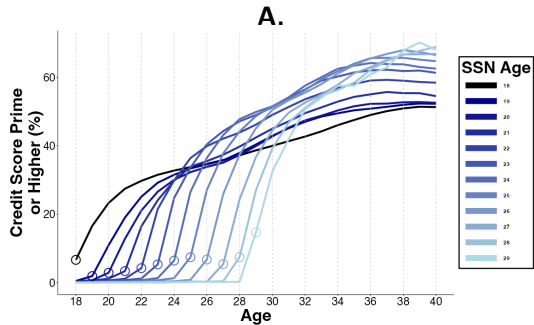
Lifecycle of Any Credit Score (Cumulative) By SSN Age Cohorts 🚀



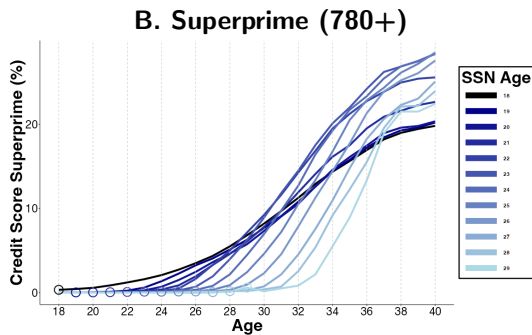
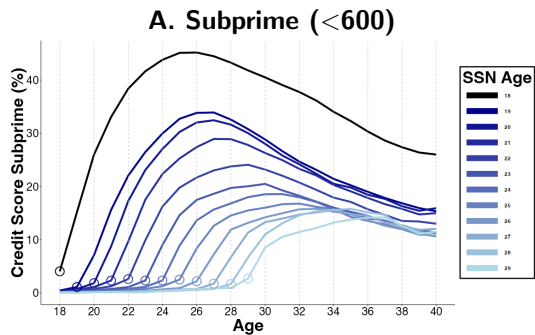
Any Credit Score At Ages 30 & 40 🗝️

Dep Var: Any Credit Score by ...	Age 30		Age 40	
	(1)	(2)	(3)	(4)
Immigrant	-13.87*** (0.16)	-0.32 (0.22)	-1.23*** (0.04)	-0.6*** (0.05)
SSN Age		-3.15*** (0.06)		-0.15*** (0.01)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.045	0.053	0.009	0.009
N	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	96.26	96.26	99.35	99.35

Prime or Higher (660+) Credit Score 🚀

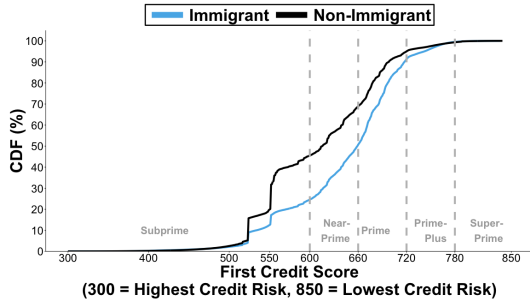


Lifecycle Of Any Subprime & Superprime Credit Scores By SSN Age 🗝️

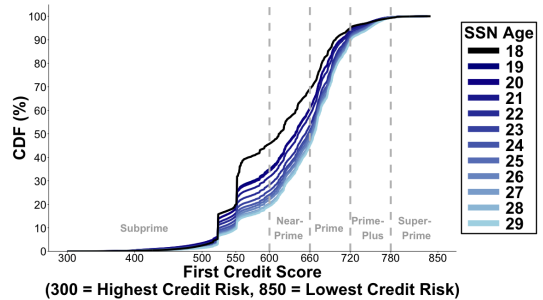


CDF Of First Credit Score By Immigration Status & SSN Age 📍

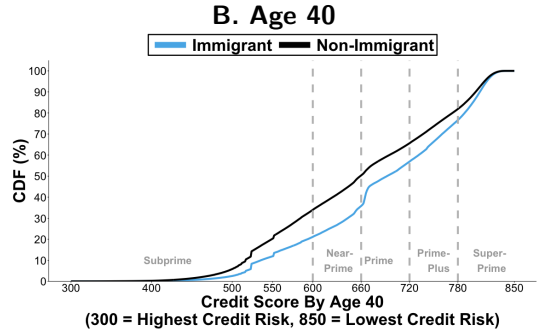
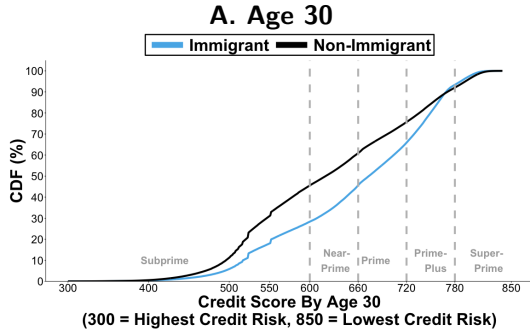
A. CDF By Immigration Status



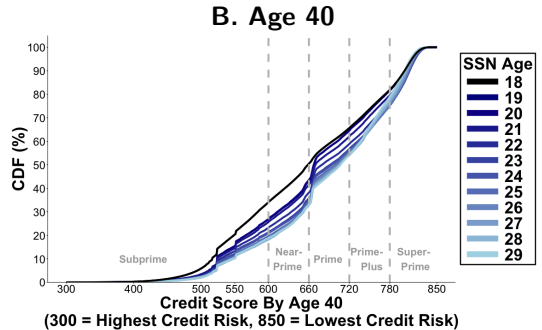
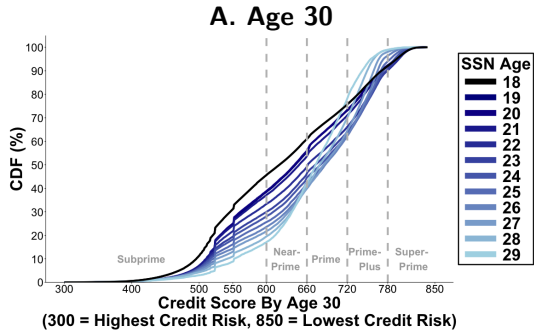
B. CDF By SSN Age



CDF Of Credit Score By Ages 30 & 40 By Immigration Status 🇺🇸



CDF Of Credit Score By Ages 30 & 40 By SSN Age 🚀



Credit Scores At Ages 30 & 40 🍷

Dep Var: Credit Score by ...	Age 30		Age 40	
	(1)	(2)	(3)	(4)
Immigrant	26.7*** (0.4)	17.2*** (0.6)	24.6*** (0.4)	14.2*** (0.5)
SSN Age		2.3*** (0.1)		2.4*** (0.1)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.135	0.135	0.124	0.124
N	6,005,430	6,005,430	6,250,733	6,250,733
Mean, SSN Age <21	624.2	624.2	658.1	658.1

Credit Score Prime or Higher (660+) At Ages 30 & 40 🗝️

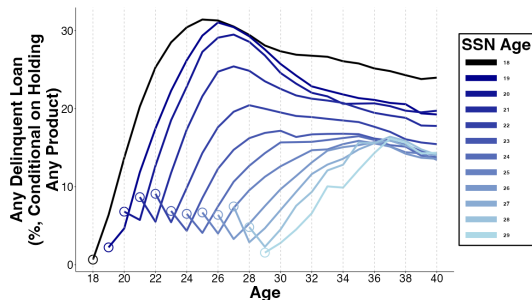
Dep Var: Prime or Higher by ...	<u>Age 30</u>		<u>Age 40</u>	
	(1)	(2)	(3)	(4)
Immigrant	4.61*** (0.17)	6.82*** (0.25)	11.24*** (0.17)	7.04*** (0.22)
SSN Age		-0.51*** (0.05)		0.98*** (0.04)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.102	0.102	0.098	0.098
N	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	37.63	37.63	49.33	49.33

Credit Scores At Ages 30 & 40, With Additional F.E. 📍

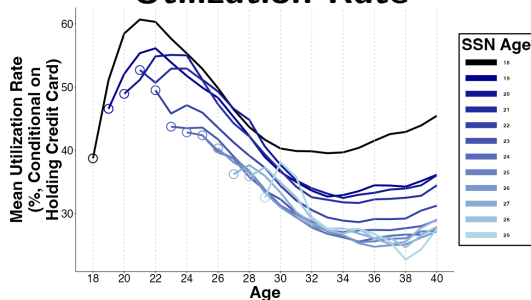
	Credit Score by...				Probability of Prime or Higher Score by...			
	Age 30	Age 40			Age 30	Age 40		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	22.21*** (0.36)	14.48*** (0.54)	22.99*** (0.34)	12.59*** (0.46)	3.78*** (0.15)	6.05*** (0.22)	10.52*** (0.14)	6.33*** (0.20)
SSN Age		1.91*** (0.10)		2.43*** (0.09)		-0.53*** (0.05)		0.98*** (0.04)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X
R^2	0.202	0.202	0.191	0.191	0.155	0.155	0.149	0.149
N	6,001,282	6,001,282	6,246,734	6,246,734	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	624.21	624.21	658.13	658.13	37.63	37.63	49.33	49.33

Immigrants (vs. Non-Immigrants) Have More Creditworthy Behaviors: Lower Delinquency Rates & Lower Credit Card Utilization

A. Any Delinquency



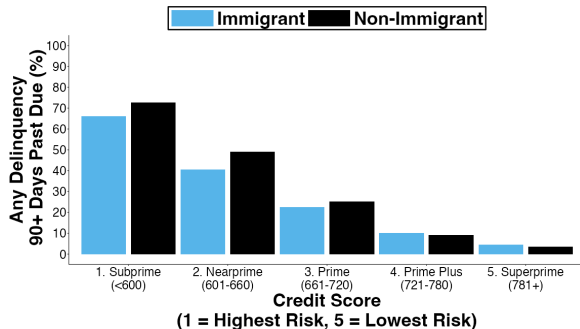
B. Mean Credit Card Utilization Rate



↓ 5% delinquency rates at age 37 conditional on credit score at age 30

Immigrants (vs. Non-Immigrants) Are More Creditworthy:

↓ 5% Delinquency Rates At Age 37 Conditional On Age 30 Credit Score 🚀



	(1)	(2)
Immigrant	-1.50*** (0.25)	-1.23*** (0.39)
SSN Age	-0.07	(0.05)
F.E. Birth Year	X	X
F.E. Age 30 Zip5	X	X
F.E. Age 30 Credit Score	X	X
R^2	0.185	0.185
N	5,804,866	5,804,866
Mean, SSN Age <21	27.18	27.18

Calibration Bias: Delinquency At 37 Conditional On Age 30 Credit Score

Dep Var: Any delinquency by age 37	<u>All</u>		<u>Subprime</u>		<u>Nearprime</u>		<u>Prime</u>		<u>Prime Plus</u>		<u>Superprime</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-1.50*** (0.10)	-1.23*** (0.16)	-2.64*** (0.21)	-3.31*** (0.36)	-3.51*** (0.23)	-2.65*** (0.41)	-1.44*** (0.16)	0.18 (0.31)	0.13 (0.09)	1.08*** (0.17)	0.29* (0.11)	0.42 (0.23)
SSN Age		-0.07* (0.03)		0.18* (0.08)		-0.20* (0.08)		-0.36*** (0.06)		-0.22*** (0.03)		-0.04 (0.06)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Zip5	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Credit Score	X	X	X	X	X	X	X	X	X	X	X	X
R^2	0.185	0.185	0.032	0.032	0.028	0.028	0.027	0.027	0.023	0.023	0.016	0.016
N	5,802,755	5,802,755	2,534,973	2,534,973	900,676	900,676	876,362	876,362	1,003,211	1,003,211	465,391	465,391
Mean, SSN Age <21	27.18	27.18	45.45	45.45	26.17	26.17	13.11	13.11	4.52	4.52	1.63	1.63

Age of Credit Products First Taken Out 🗝️

Dep Var: Age at First...	<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	4.96*** (0.05)	2.27*** (0.05)	2.26*** (0.04)	0.40*** (0.04)	4.80*** (0.02)	1.79*** (0.03)
SSN Age		0.63*** (0.01)		0.43*** (0.01)		0.70*** (0.01)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.083	0.084	0.082	0.083	0.082	0.086
N	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	29.40	29.40	36.49	36.49	22.99	22.99

Age of Credit Products First Taken Out (With Additional F.E.)

Dep Var: Age at First...	<u>Credit Report</u>		<u>Credit Product</u>		<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	5.15*** (0.02)	2.07*** (0.02)	4.73*** (0.02)	1.81*** (0.03)	3.88*** (0.04)	1.60*** (0.04)	1.72*** (0.03)	0.03 (0.04)	4.21*** (0.02)	1.49*** (0.03)
SSN Age		0.72*** (0.01)		0.68*** (0.01)		0.53*** (0.01)		0.39*** (0.01)		0.63*** (0.01)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X	X	X
R^2	0.260	0.275	0.200	0.207	0.180	0.181	0.158	0.159	0.173	0.176
N	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	20.19	20.19	21.56	21.56	29.40	29.40	36.49	36.49	22.99	22.99

Credit Access By Age 37

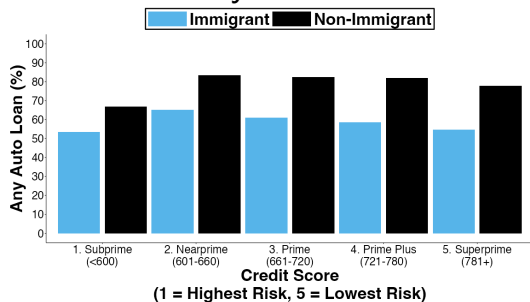
Dep. Var: By Age 37, has ...	<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-15.66*** (0.18)	-8.69*** (0.23)	-8.91*** (0.17)	-1.05*** (0.23)	-1.35*** (0.08)	-0.37*** (0.11)
SSN Age		-1.62*** (0.04)		-1.83*** (0.04)		-0.23*** (0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.041	0.042	0.063	0.063	0.024	0.024
N	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	76.06	76.06	44.63	44.63	94.12	94.12

Credit Access By Age 37 With Additional F.E.

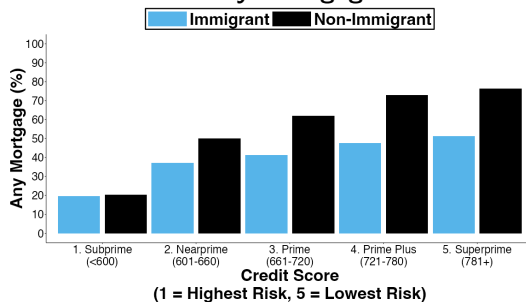
Dep Var: By Age 37, has...	<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-10.61*** (0.17)	-5.62*** (0.21)	-5.21*** (0.17)	1.39*** (0.21)	0.09 (0.07)	0.37*** (0.10)
SSN Age		-1.16*** (0.04)		-1.54*** (0.04)		-0.06*** (0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.137	0.137	0.151	0.151	0.070	0.070
N	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	76.06	76.06	44.63	44.63	94.12	94.12

Any Auto Loan & Mortgage By Age 37 Conditional On Age 30 Credit Score 📍

A. Any Auto Loan



B. Any Mortgage

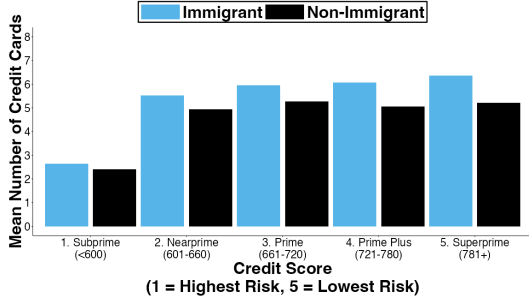


Age 37 Credit Outcomes Conditional On Age 30 Credit Score 🍷

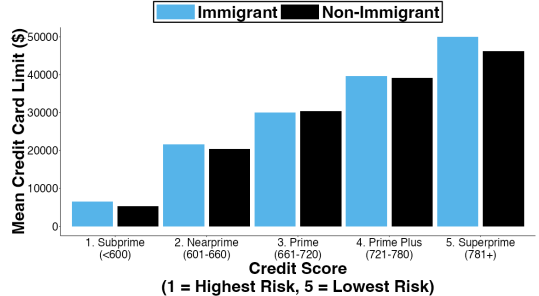
Dep Var: Outcomes at Age 37...	<u>Any</u> <u>Auto Loan</u>		<u>Any</u> <u>Mortgage</u>		<u>Any</u> <u>Credit Card</u>		<u>Number of</u> <u>Credit Cards</u>		<u>Credit</u> <u>Card Limits</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	-15.01*** (0.20)	-9.50*** (0.24)	-12.05*** (0.20)	-2.71*** (0.25)	-1.55*** (0.08)	-1.85*** (0.12)	0.36*** (0.01)	0.17*** (0.02)	-937.9*** (147.1)	732.3*** (179.4)
SSN Age		-1.35*** (0.04)		-2.29*** (0.05)		0.07*** (0.02)		0.05*** (0.00)		-409.3*** (35.1)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Credit Score	X	X	X	X	X	X	X	X	X	X
R^2	0.108	0.108	0.267	0.268	0.110	0.110	0.141	0.141	0.305	0.305
N	5,802,755	5,802,755	5,802,755	5,802,755	5,802,755	5,802,755	5,802,755	5802755	5,802,755	5,802,755
Mean, SSN Age <21	75.1	75.1	44.54	44.54	92.06	92.06	3.9	3.9	\$20,431.9	\$20,431.9

Credit Cards By Age 37 Conditional On Age 30 Credit Score 📍

A. Number of Credit Cards

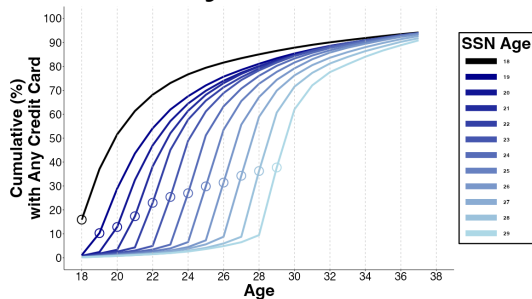


B. Credit Card Limits

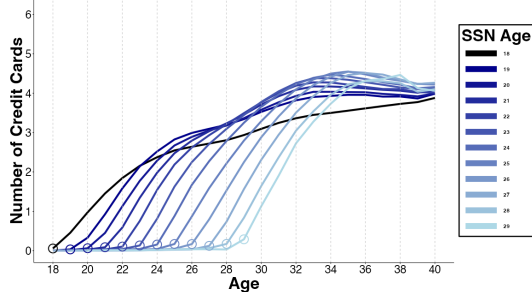


Immigrants' Access to Credit Cards Converges With Non-Immigrants

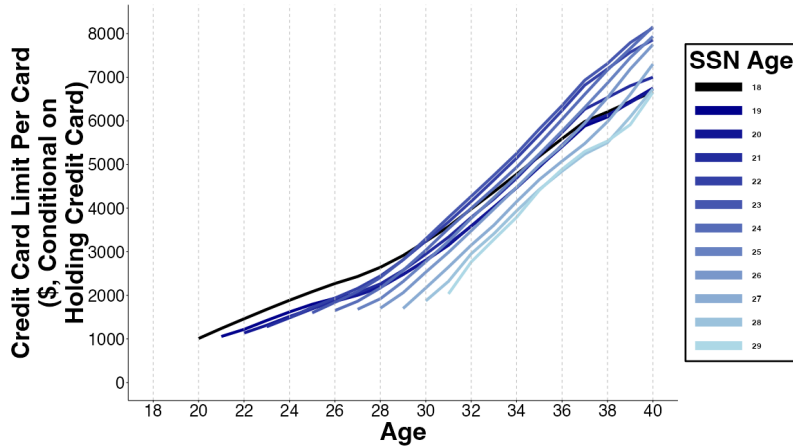
A. Any Credit Card



B. Number of Credit Cards

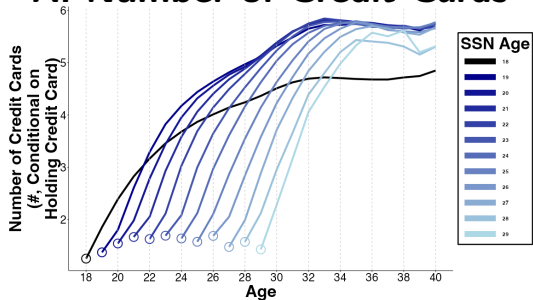


Credit Card Limit Per Card 🚀

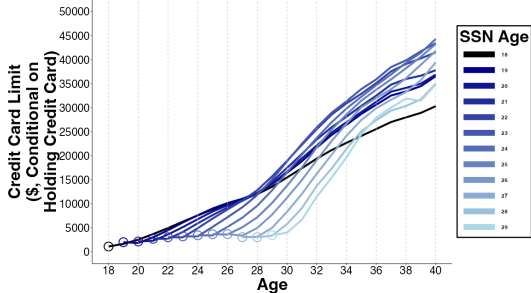


Immigrants' Credit Card Access Conditional On Holding Any Credit Card

A. Number of Credit Cards



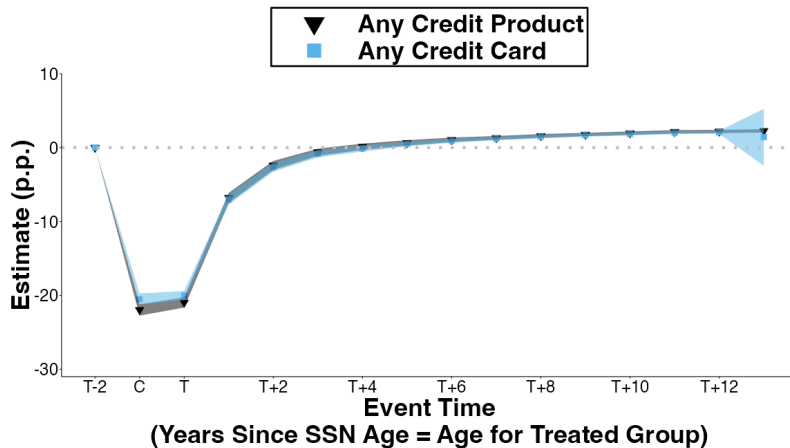
B. Credit Card Limits



Credit Card Outcomes At Age 30 & At Age 40 🐦

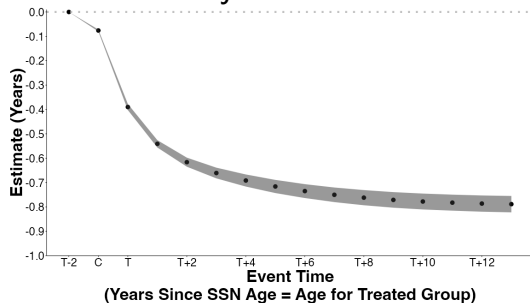
	Number of Credit Cards At...				Credit Card Limits At...			
	Age 30		Age 40		Age 30		Age 40	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-0.13*** (0.02)	1.12*** (0.02)	0.57*** (0.02)	0.15*** (0.02)	-2,748.6*** (98.0)	3,027.2*** (90.8)	2,296.3*** (140.5)	1,539.3*** (167.5)
SSN Age		-0.29*** (0.00)		0.09*** (0.00)		-1,345.2*** (21.0)		168.5*** (32.0)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X
R^2	0.126	0.128	0.096	0.096	0.141	0.143	0.182	0.182
N	6,293,372	6,293,372	4,914,884	4,914,884	6,293,372	6,293,372	4,914,884	4,914,884
Mean, SSN Age <21	3.35	3.35	3.92	3.92	\$11,491.7	\$11,491.7	\$22,112.0	\$22,112.0

Paired Cohorts: First Credit Product & First Credit Card 🚀

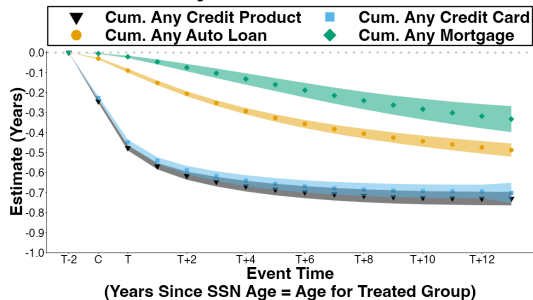


Paired Cohorts: Cumulatives 🚀

A. Any Credit Score



B. By Credit Product

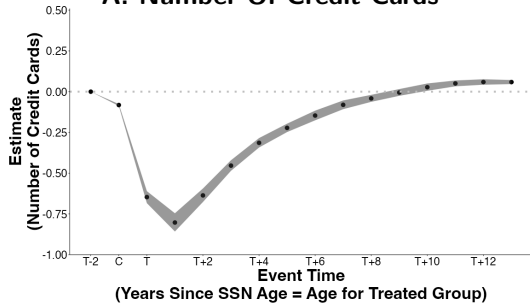


Paired Cohorts: Cumulative Years of Delay 🚀

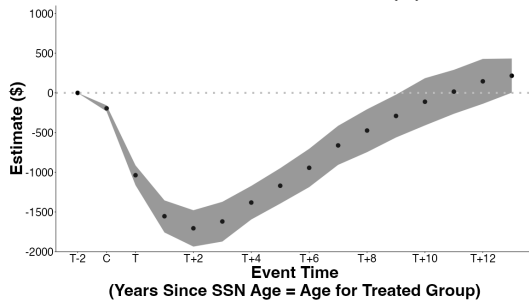
Dep. Var: Any ...	<u>Credit Score</u> (1)	<u>Credit Product</u> (2)	<u>Auto Loan</u> (3)	<u>Mortgage</u> (4)	<u>Credit Card</u> (5)
<i>C</i>	-0.076*** [-0.080, -0.072]	-0.243*** [-0.251, -0.235]	-0.030*** [-0.034, -0.026]	-0.005*** [-0.006, -0.003]	-0.227*** [-0.235, -0.218]
<i>T</i>	-0.390*** [-0.403, -0.377]	-0.476*** [-0.490, -0.463]	-0.090*** [-0.100, -0.080]	-0.021*** [-0.027, -0.015]	-0.448*** [-0.463, -0.434]
<i>T + 5</i>	-0.716*** [-0.745, -0.686]	-0.684*** [-0.717, -0.653]	-0.327*** [-0.347, -0.308]	-0.160*** [-0.195, -0.126]	-0.656*** [-0.689, -0.625]
<i>T + 10</i>	-0.778*** [-0.813, -0.741]	-0.725*** [-0.762, -0.689]	-0.443*** [-0.473, -0.412]	-0.283*** [-0.338, -0.228]	-0.692*** [-0.729, -0.656]
<i>N</i> Paired Cohorts	68	68	68	68	68
<i>N</i> Consumers	463,542	463,542	463,542	463,542	463,542
<i>N</i>	7,416,672	7,416,672	7,416,672	7,416,672	7,416,672

Paired Cohorts: Credit Cards 🚀

A. Number Of Credit Cards



B. Credit Card Limits (\$)



Potential Mechanism: Emigration (Return Migration)

Proxy emigration by **attrition** from data

Immigrants exit data more than Non-Immigrants

- If arrive earlier, on average, exit data earlier, so can't explain paired cohort results

Restrict to consumers observed in 2024:

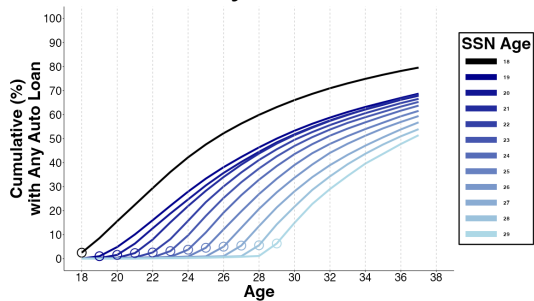
- **Mortgage** Immigrant vs. Non-Immigrant gap closes but SSN Age gap remains
- **Auto loan** Immigrant vs. Non-Immigrant gap & SSN Age gap remains

Consumers Observed In Data In 2024 🇺🇸

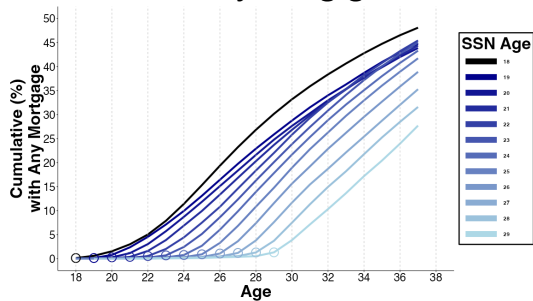
Dep Var: In 2024, has...	Any Credit Report		Any Credit Score		Any Credit Tradeline	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-2.10*** (0.10)	-4.38*** (0.16)	-6.25*** (0.18)	-11.19*** (0.24)	-7.39*** (0.21)	-11.83*** (0.25)
SSN Age		0.53*** (0.02)		1.15*** (0.04)		1.04*** (0.04)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.045	0.045	0.112	0.113	0.147	0.147
N	6,293,372	6,293,372	6293372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	95.92	95.92	89.94	89.94	84.34	84.34

Lifecycle Of Credit For Consumers Observed In 2024 🗺️

A. Any Auto Loan



B. Any Mortgage



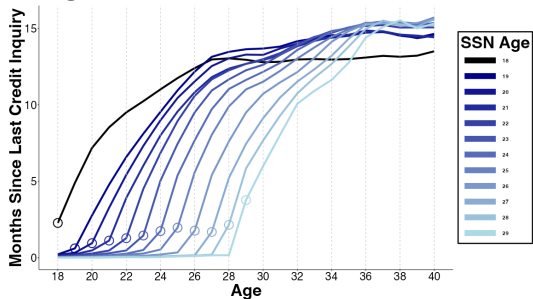
Credit Access By Age 37 With Additional F.E. Conditional On Any Credit Score Observed in 2024 📍

Dep Var: By Age 37, has...	<u>Auto Loan</u>		<u>Mortgage</u>		<u>Credit Card</u>	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-10.75*** (0.17)	-3.98*** (0.22)	-4.48*** (0.18)	4.88*** (0.23)	-0.49*** (0.07)	0.14 (0.09)
SSN Age		-1.56*** (0.04)		-2.15*** (0.05)		-0.14*** (0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.11	0.111	0.138	0.138	0.061	0.061
N	5,624,238	5,624,238	5,624,238	5,624,238	5,624,238	5,624,238
Mean, SSN Age <21	79.38	79.38	48.03	48.03	95.38	95.38

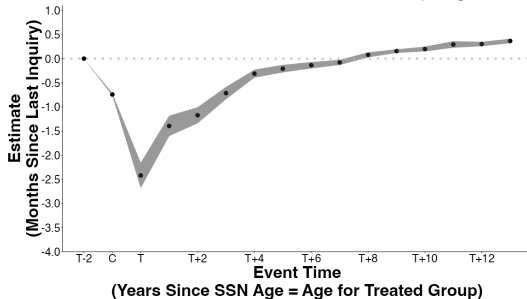
Does Lower Credit Demand Explain Lower Access?

Immigrants Have $\uparrow$ Credit Demand In 20s, $\downarrow$ Demand In 30s 🚀

A. Months Since Last Credit Inquiry
Higher Value = Lower Credit Demand



B. Paired Cohorts
Months Since Last Credit Inquiry



Caveat: Lower credit demand in 30s may be function of earlier rejections $\rightarrow$ deterrence

Not Clear Lower Immigrant Demand for Debt (SCF, 2022)

...But Low Statistical Power To Detect Out Differences

Dep Var: Apply For...	<u>Any Credit</u>		<u>Auto</u>		<u>Mortgage</u>		<u>Credit Card</u>		<u>Other Credit</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	2.32 (4.56)	-1.08 (7.17)	2.03 (4.37)	-6.55 (7.53)	2.11 (3.80)	5.65 (6.85)	1.63 (3.22)	2.02 (5.32)	1.84 (3.62)	-7.66 ⁺ (4.61)
Immigration Age		0.44 (0.62)		1.10 (0.72)		-0.45 (0.65)		-0.05 (0.48)		1.22* (0.55)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.051	0.051	0.051	0.053	0.045	0.045	0.045	0.045	0.026	0.029
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age <21	64.81	64.81	19.95	19.95	11.28	11.28	9.28	9.28	12.08	12.08

*** 0.5%, ** 1%, * 5%, + 10% statistical significance

Lower Incomes for Immigrants & Later-in-Life Immigrants (5-year ACS waves)

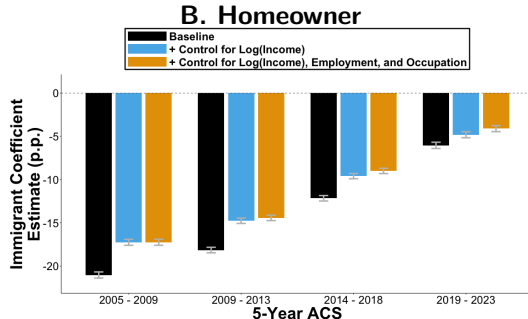
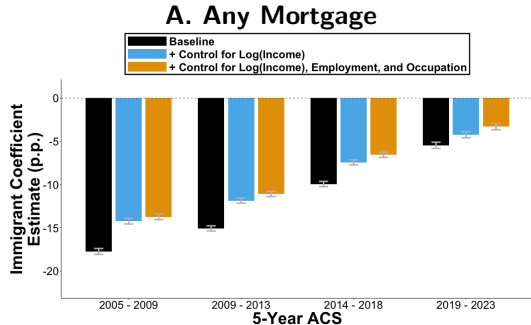
	Log (Income)							
	2005 - 2009		2009 - 2013		2014 - 2018		2019 - 2023	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-17.66*** (0.36)	-11.79*** (0.67)	-17.41*** (0.31)	-10.18*** (0.56)	-14.51*** (0.28)	-8.55*** (0.53)	-8.08*** (0.34)	-3.03*** (0.64)
Immigration Age		-1.49*** (0.14)		-1.69*** (0.11)		-1.27*** (0.10)		-1.06*** (0.12)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. Microdata Area	X	X	X	X	X	X	X	X
F.E. Sex	X	X	X	X	X	X	X	X
F.E. Education	X	X	X	X	X	X	X	X
F.E. Race	X	X	X	X	X	X	X	X
F.E. Hispanic	X	X	X	X	X	X	X	X
F.E. Survey Year	X	X	X	X	X	X	X	X
N	2,079,500	2,079,500	2,162,012	2,162,012	2,231,275	2,231,275	2,200,018	2,200,018
R ²	0.161	0.161	0.196	0.196	0.239	0.239	0.228	0.228
Mean Income, Immigration Age <21	\$71,088	\$71,088	\$78,339	\$78,339	\$98,660	\$98,660	\$134,099	\$134,099

- Consistent with Chiswick (1978); Lubotsky (2007); Kerr, Kerr, & Smith (2026)
- Robust to additional F.E. for employment & occupation

Income, Controlling for Demographics, Employment & Occupation (ACS)

	Log (Income)							
	2005 - 2009		2009 - 2013		2014 - 2018		2019 - 2023	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-11.23*** (0.36)	-7.57*** (0.65)	-10.98*** (0.30)	-6.50*** (0.54)	-9.15*** (0.27)	-5.43*** (0.51)	-4.41*** (0.33)	-1.27* (0.61)
Immigration Age		-0.93*** (0.14)		-1.05*** (0.11)		-0.79*** (0.09)		-0.66*** (0.11)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. Microdata Area	X	X	X	X	X	X	X	X
F.E. Sex	X	X	X	X	X	X	X	X
F.E. Education	X	X	X	X	X	X	X	X
F.E. Race	X	X	X	X	X	X	X	X
F.E. Hispanic	X	X	X	X	X	X	X	X
F.E. Survey Year	X	X	X	X	X	X	X	X
F.E. Employment	X	X	X	X	X	X	X	X
F.E. Occupation	X	X	X	X	X	X	X	X
N	2,079,500	2,079,500	2,162,012	2,162,012	2,231,275	2,231,275	2,200,018	2,200,018
R ²	0.199	0.199	0.248	0.248	0.299	0.299	0.299	0.299
Mean Income, Immigration Age <21	\$71,088	\$71,088	\$78,339	\$78,339	\$98,660	\$98,660	\$134,099	\$134,099

Differences in Mortgage & Homeownership Remain After Controlling for Income & Demographics (5-year ACS waves)



Immigrating one year later in life leads to persistently lower mortgage use & homeownership

All regressions include fixed effects for...

birth year, geographic area, sex, education levels, race, Hispanic, & survey year

Use 2022 Survey of Consumer Finances (SCF) To Evaluate Mechanisms

SCF 2022 includes if Immigrant & Age of Immigration

Adapt earlier cross-sectional regression to this data:

$$Y_i = \beta_1 \cdot \mathbb{I}\{\text{Immigrant}\}_i + \beta_2 \cdot \text{Immigration Age}_i + \mu_{b(i)} + \eta_{d(i)} + \epsilon_i$$

$\eta_{d(i)}$ includes control for Log(Income) and F.E.: male, race, Hispanic, spouse, marital status, household size, number of children, & education levels

SCF Debt Attitude Questions:

“People have many different reasons for borrowing money which they pay back over a period of time. For each of the reasons I read, please tell me whether you feel it is all right for someone like yourself to borrow money...to finance the purchase of a car.”

Debt Attitudes (SCF, 2022) 📌

Dep Var:	<u>Auto Debt Attitude</u>		<u>Any Auto Debt</u>		<u>Any Auto Debt</u>		<u>General Debt Attitude</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-15.41*** (4.58)	-23.02*** (7.25)	-4.27 (5.08)	-11.14 (8.27)	-0.21 (5.01)	-5.10 (7.87)	-8.96 (7.92)	-15.94 (13.71)
Immigration Age		0.98 (0.62)		0.88 (0.76)		0.63 (0.69)		0.89 (1.19)
Auto Debt Attitude					26.34*** (3.23)	26.24*** (3.24)		
Demographic Controls	X	X	X	X	X	X	X	X
R^2	0.049	0.050	0.063	0.063	0.106	0.106	0.026	0.026
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age <21	79.58	79.58	47.14	47.14	47.14	47.14	3.80	3.80

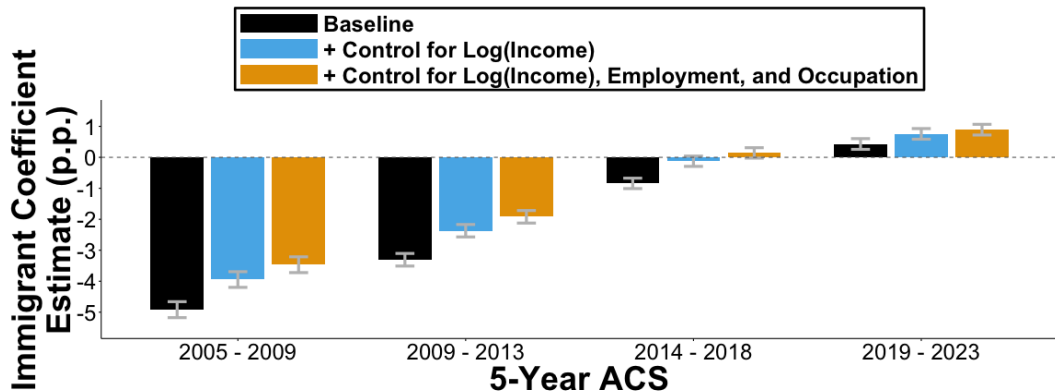
*** 0.5%, ** 1%, * 5%, + 10% statistical significance

More Granular Debt Attitudes (SCF, 2022) 📌

Dep Var: All Right To Borrow...	<u>In General (Positive)</u>		<u>In General (Negative)</u>		<u>For Living Expenses</u>		<u>For Vacation</u>		<u>For Education</u>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	-4.27 (4.83)	-4.29 (8.95)	4.69 (4.65)	11.65 (7.09)	-8.76 ⁺ (4.89)	-8.35 (7.66)	-5.01 (3.40)	-8.37 (5.21)	-2.80 (4.18)	9.07 (5.59)
Immigration Age		0.00 (0.80)		-0.89 (0.62)		-0.05 (0.74)		0.43 (0.56)		-1.52* (0.67)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.016	0.016	0.043	0.044	0.032	0.032	0.020	0.020	0.040	0.045
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1813
Mean, Immigration Age <21	29.51	29.51	25.71	25.71	73.38	73.38	15.30	15.30	84.24	84.24

*** 0.5%, ** 1%, * 5%, + 10% statistical significance

Car Ownership Difference Disappears (5-year ACS waves)



All regressions include fixed effects for...

birth year, geographic area, sex, education levels, race, Hispanic, & survey year

Access to mortgage, homeownership, and car ownership (ACS)

	2005 - 2009			2009 - 2013			2014 - 2018			2019 - 2023		
A. Dep Var: Any Mortgage	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-9.90*** (0.31)	-7.57*** (0.30)	-6.90*** (0.30)	-8.09*** (0.28)	-6.21*** (0.27)	-5.38*** (0.27)	-3.56*** (0.29)	-2.09*** (0.28)	-1.23*** (0.28)	-1.86*** (0.35)	-1.40*** (0.34)	-0.65* (0.34)
Immigration Age	-1.98*** (0.06)	-1.69*** (0.06)	-1.73*** (0.06)	-1.63*** (0.05)	-1.32*** (0.05)	-1.34*** (0.05)	-1.36*** (0.05)	-1.14*** (0.05)	-1.13*** (0.05)	-0.76*** (0.06)	-0.60*** (0.06)	-0.56*** (0.06)
Log (Income)		17.92*** (0.05)	18.13*** (0.05)		16.43*** (0.05)	16.41*** (0.05)		15.23*** (0.05)	14.88*** (0.05)		13.88*** (0.06)	13.12*** (0.06)
R ²	0.100	0.213	0.220	0.116	0.205	0.212	0.135	0.206	0.213	0.131	0.192	0.199
Mean, Immigration Age <21	44.22	44.22	44.22	44.65	44.65	44.65	48.12	48.12	48.12	54.25	54.25	54.25
B. Dep Var: Homeowner	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-13.08*** (0.32)	-10.56*** (0.32)	-10.19*** (0.32)	-10.65*** (0.30)	-8.67*** (0.29)	-8.04*** (0.29)	-4.82*** (0.30)	-3.31*** (0.29)	-2.50*** (0.29)	-0.83** (0.35)	-0.37 (0.34)	0.35 (0.34)
Immigration Age	-2.02*** (0.07)	-1.70*** (0.07)	-1.79*** (0.06)	-1.75*** (0.06)	-1.42*** (0.06)	-1.50*** (0.06)	-1.56*** (0.05)	-1.34*** (0.05)	-1.38*** (0.05)	-1.10*** (0.06)	-0.93*** (0.06)	-0.94*** (0.06)
Log (Income)		19.37*** (0.05)	19.91*** (0.05)		17.31*** (0.05)	17.78*** (0.05)		15.62*** (0.05)	15.86*** (0.05)		13.80*** (0.06)	13.71*** (0.06)
R ²	0.114	0.244	0.255	0.128	0.225	0.235	0.152	0.229	0.237	0.155	0.224	0.233
Mean, Immigration Age <21	52.40	52.40	52.40	53.25	53.25	53.25	59.26	59.26	59.26	68.54	68.54	68.54
C. Dep Var: Any Car	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-2.37*** (0.25)	-1.72*** (0.24)	-1.34*** (0.24)	-0.88*** (0.20)	-0.34* (0.19)	0.00 (0.19)	1.11*** (0.16)	1.53*** (0.16)	1.69*** (0.16)	1.18*** (0.17)	1.30*** (0.17)	1.39*** (0.17)
Immigration Age	-0.65*** (0.05)	-0.56*** (0.05)	-0.54*** (0.05)	-0.57*** (0.04)	-0.48*** (0.04)	-0.45*** (0.04)	-0.41*** (0.03)	-0.35*** (0.03)	-0.33*** (0.03)	-0.16*** (0.03)	-0.11*** (0.03)	-0.10*** (0.03)
Log (Income)		4.99*** (0.03)	4.85*** (0.03)		4.80*** (0.03)	4.61*** (0.03)		4.36*** (0.03)	4.15*** (0.03)		3.67*** (0.03)	3.48*** (0.03)
R ²	0.200	0.234	0.236	0.211	0.240	0.243	0.187	0.215	0.219	0.153	0.177	0.181
Mean, Immigration Age <21	93.51	93.51	93.51	93.66	93.66	93.66	94.69	94.69	94.69	95.40	95.40	95.40